Commercialization Strategy: Retinal Disease AI Platform

Classification: Internal Strategy Document
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System: Multi-label retinal disease classification (45 diseases) with clinical knowledge graph reasoning

1. Executive Summary

This document outlines the commercialization strategy for a production-grade AI-powered retinal disease screening platform that classifies 45 retinal conditions from fundus images using novel Graph Neural Network architectures with embedded clinical knowledge graphs.

The core thesis: Ophthalmology faces a global capacity crisis — there are ~232,000 ophthalmologists serving 2.2 billion people with vision impairment worldwide. Multi-label AI screening that detects 45 conditions simultaneously (not just single-disease like most competitors) can compress a 20-minute specialist review into a 25-millisecond inference, unlocking screening at population scale — especially in Sub-Saharan Africa and South/Southeast Asia where the ophthalmologist-to-population ratio can exceed 1:1,000,000.

Strategic positioning: Not a replacement for the ophthalmologist, but a force multiplier — a clinical decision support system (CDSS) that triages, prioritizes, and surfaces the 15-20% of scans that need urgent specialist attention, while safely clearing the 80% that are normal or low-risk.

Key Numbers

Metric	Value
Total addressable market (TAM)	$5.4B (ophthalmic AI, 2026)
Serviceable addressable market (SAM)	$1.8B (multi-disease screening)
Serviceable obtainable market (SOM)	$45M (Year 3 target)
Unit economics target	$0.30-0.80 per scan (volume-dependent)
Regulatory timeline	CE Mark Q4 2026, FDA De Novo Q3 2027
Break-even target	Month 18 post-launch

2. Market Analysis: Ophthalmic AI in 2026

2.1 Market Size & Growth

The global ophthalmic AI market is valued at approximately **$5.4B in 2026**, growing at **31.2% CAGR** through 2031. Key drivers:

- **Diabetes epidemic**: 537M adults with diabetes globally (IDF 2024); ~35% develop diabetic retinopathy
- **Aging populations**: Age-related macular degeneration affects 196M people globally
- **Screening mandates**: WHO 2025 resolution calling for universal eye screening in primary care
- **Workforce shortage**: Demand for eye exams growing 3x faster than ophthalmologist supply

2.2 Market Segmentation

Segment	Description	Size (2026)	Growth
Diabetic retinopathy screening	Single-disease, high-volume primary care	$2.1B	28% CAGR
Multi-disease screening	Multi-label detection in specialty clinics	$1.8B	36% CAGR
Surgical planning AI	Pre-operative analysis	$0.8B	25% CAGR
Remote/teleophthalmology	Cloud-based screening for rural areas	$0.7B	42% CAGR

Our primary segment: Multi-disease screening ($1.8B) + Remote/teleophthalmology ($0.7B) = **$2.5B combined TAM**.

2.3 Buyer Personas

Persona	Role	Pain Point	Decision Criteria
Chief Medical Officer	Clinical champion	Quality assurance, patient safety	Clinical validation, explainability, regulatory status
CIO / CTO	Technical buyer	Integration complexity, security	API-first architecture, deployment flexibility, SOC2
Ophthalmologist	End user	Screening backlog, alert fatigue	Speed, accuracy, referral prioritization, trust
Healthcare Administrator	Budget holder	Cost per screening, throughput	ROI, reimbursement codes, operational efficiency
Ministry of Health (LMIC)	Population health buyer	Specialist shortage, rural coverage	Offline capability, cost per scan, language support

3. Competitive Landscape & Positioning

3.1 Competitor Matrix

Company	Diseases	Modality	Regulatory	Pricing	Weakness
IDx-DR (Digital Diagnostics)	DR only (binary)	Fundus	FDA cleared (2018)	~$40/test	Single disease, requires Topcon camera
EyeArt (Eyenuk)	DR + DME	Fundus	FDA cleared (2020)	$25-35/test	2 diseases only, no explainability
RetinAI	DR, AMD, Glaucoma	OCT + Fundus	CE Mark	Enterprise license	OCT-dependent, expensive hardware
Google Health (ARDA)	DR, DME	Fundus	CE Mark (Thailand)	Research/ partnership	Not commercially available, single disease
Visulytix (Pegasus)	DR, AMD	Fundus	CE Mark (UK)	~$15/test	2 diseases, limited geography
Our Platform	**45 diseases**	Fundus	Pre-conformity	**$0.30-0.80/ scan**	Pre-regulatory, dataset scale

3.2 Competitive Moats

Our differentiation rests on five defensible advantages:

1. Multi-Label Breadth (45 diseases)
Every competitor screens for 1-3 diseases. We detect 45 simultaneously from a single fundus image. This means one scan replaces what would otherwise require multiple specialist consultations. The clinical knowledge graph encodes 144 disease co-occurrence relationships (e.g., DR often co-presents with HR, BRVO), enabling clinically coherent multi-label predictions.

2. Clinical Knowledge Graph Reasoning
Our ClinicalKnowledgeGraph is not just a classifier — it embeds domain knowledge about disease prevalence, severity hierarchies, and co-occurrence patterns directly into inference. This produces clinically plausible predictions, reduces impossible label combinations, and generates referral priority rankings (Emergency / Urgent / Routine).

3. Four Explainability Methods
GradCAM, LIME, SHAP, and Integrated Gradients are all production-ready. In 2026, regulatory bodies (FDA, EU AI Act) increasingly require explainability for high-risk AI. Most competitors offer none or one method. We ship four, giving clinicians choice in how they verify AI reasoning.

4. Cost Structure Advantage
25ms inference latency means one GPU can serve ~40 scans/second. At $0.30-0.80/scan, we undercut competitors by 30-50x while screening for 15-45x more conditions. This cost advantage is structural — GNN architectures are inherently more parameter-efficient than the large vision transformers competitors use.

5. LMIC-First Design
CPU inference support, Docker-based deployment, and clinical knowledge graph calibrated with Ugandan disease prevalence data. This is not an afterthought — the system was designed for deployment in resource-constrained settings from day one.

3.3 Positioning Statement

> *For ophthalmologists and health systems overwhelmed by screening backlogs, our platform is the only AI clinical decision support system that detects 45 retinal diseases simultaneously from a single fundus image at sub-$1 per scan, with built-in clinical reasoning and four explainability methods — enabling population-scale screening without population-scale ophthalmologist headcount.*

4. Business Model Architecture

4.1 Business Model Canvas

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│                        BUSINESS MODEL
CANVAS                                │
├──────────────┬──────────────┬───────────────┬──────────────┬─────────────
───┤
│ KEY PARTNERS │ KEY          │ VALUE         │ CUSTOMER     │
CUSTOMER        │
│              │ ACTIVITIES   │ PROPOSITIONS  │ RELATIONS    │
SEGMENTS        │
│              │              │               │
│                 │
│ • Fundus     │ • Clinical   │ • 45-disease  │ • Clinical   │ •
Hospital      │
│   camera OEMs│   validation │   screening   │   success    │
networks      │
│   (Topcon,   │   studies    │   in one scan │   managers   │   (Tier
1)      │
│   Canon,     │ • Regulatory │ • Sub-$1/scan │ • 24/7       │ •
Specialty     │
│   Optomed)   │   submissions│ • Explainable │   clinical   │   eye clinic
s   │
│ • Regulatory │ • Model      │   AI (4       │   support    │   (Tier
2)      │
│   consultants│   retraining │   methods)    │ • Quarterly  │ •
Public        │
│   (FDA, CE)  │ • Customer   │ • Referral    │   model      │
health /      │
│ • CROs for   │   onboarding │   priority    │   performance│
Ministries    │
│   clinical   │ • Platform   │   ranking     │   reviews    │   of
Health     │
│   trials     │   ops + SRE  │ • EU AI Act   │ • Training   │   (Tier
3)      │
│ • Cloud      │              │   ready       │   workshops  │ •
Telehealth    │
│   (AWS/GCP)  │              │ • CPU + GPU   │              │
platforms     │
│ • Research   │              │   deployment  │              │   (Tier
4)      │
│   hospitals  │              │               │
│                 │
├──────────────┴──────────────┼───────────────┼──────────────┴─────────────
───┤
│ KEY RESOURCES               │               │
CHANNELS                    │
│                             │
│                               │
│ • 4 GNN model architectures │               │ • Direct sales
(enterprise)   │
│ • Clinical knowledge graph  │               │ • Camera OEM
partnerships      │
│   (144 relationships)       │               │ • Distributor network
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(LMIC)  │
│ • MLOps infrastructure      │               │ • Medical conferences
(AAO,   │
│ • Regulatory dossier        │               │   ARVO, EURETINA,
WOC)        │
│ • Clinical validation data  │               │ • Published clinical studie
s  │
│ • Engineering team          │               │ • Government tenders
(LMIC)   │
├─────────────────────────────┴───────────────┴────────────────────────────
───┤
│ COST STRUCTURE                              │ REVENUE
STREAMS                │
│
│                               │
│ Fixed:                                      │ • Per-scan fees
($0.30-0.80)  │
│ • Engineering team (8-12 FTE): $1.2M/yr     │ • Annual platform
licenses    │
│ • Regulatory (FDA + CE): $400K one-time     │ • Enterprise SaaS
contracts   │
│ • Clinical validation: $300K/study          │ • Government/NGO
screening    │
│ • Cloud infrastructure: $180K/yr            │   programs (volume
pricing)   │
│ • Regulatory maintenance: $100K/yr          │ • OEM licensing
(camera       │
│                                             │
integration)                │
│ Variable:                                   │ • Training &
certification    │
│ • GPU compute: ~$0.02/scan (at scale)       │ • Data insights &
analytics   │
│ • Customer success: $50K/enterprise/yr
│                               │
│ • Regulatory per-market: $80K-200K
│                               │
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4.2 Revenue Model: Tiered Architecture

We employ a **hybrid revenue model** combining per-scan transactional pricing with platform subscription fees, structured across four tiers:

Tier 1: Hospital Networks (Enterprise)

Component	Pricing	Notes
Platform license	$48,000-120,000/year	Based on # locations
Per-scan fee	$0.50-0.80	Decreasing with volume
Implementation	$15,000-30,000 one-time	Integration, training, validation
Clinical support	$24,000/year	Dedicated clinical success manager
SLA guarantee	Included	p99 < 100ms, 99.9% uptime

Target: 20-50 hospital networks by Year 3
ACV: $80,000-$180,000

Tier 2: Specialty Eye Clinics (SMB)

Component	Pricing	Notes
Monthly subscription	$1,500-3,500/month	Includes up to 2,000 scans
Overage per-scan	$0.60	Above included volume
Onboarding	$5,000 one-time	Remote setup + training

Target: 100-300 clinics by Year 3
ACV: $24,000-$48,000

Tier 3: Public Health / Government Programs (Volume)

Component	Pricing	Notes
Per-scan fee	$0.30-0.50	Volume-based (>100K scans/year)
Platform deployment	$25,000-50,000/year	On-premise or sovereign cloud
Training program	$10,000/cohort	Train-the-trainer model

Target: 5-15 government programs by Year 3
ACV: $60,000-$300,000 (highly variable by population)

Tier 4: OEM / Platform Integration

Component	Pricing	Notes
API access license	$100,000-500,000/year	White-label or co-branded
Per-scan royalty	$0.15-0.30	Embedded in partner pricing
Integration support	$50,000 one-time	SDK, documentation, testing

Target: 3-8 OEM partners by Year 3 (camera manufacturers, telehealth platforms, EHR vendors)

4.3 Unit Economics

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Revenue per scan (blended):                    $0.55
├── GPU compute cost:                         -$0.02
├── Cloud infrastructure (amortized):         -$0.03
├── Customer success (amortized):             -$0.05
├── Regulatory maintenance (amortized):       -$0.02
├── Model retraining (amortized):             -$0.01
└── Gross margin per scan:                     $0.42 (76.4%)

Contribution margin after S&M (amortized):     $0.32 (58.2%)

Break-even volume:    ~4.8M scans/year (or equivalent in platform fees)
Break-even timeline:  Month 18 post-launch (with Tier 1+2 pipeline)
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5. Regulatory Strategy

5.1 Classification & Pathway

Jurisdiction	Classification	Pathway	Timeline	Cost
EU (CE Mark)	Class IIa Medical Device + High-Risk AI	MDR 2017/745 + EU AI Act conformity	Q4 2026 - Q1 2027	$150K-250K
FDA (USA)	Class II SaMD (Computer-Aided Detection)	De Novo (no predicate for 45-disease)	Q2 2027 - Q4 2027	$200K-350K
EAC (East Africa)	Medical device (varies by country)	National registration (Uganda, Kenya, Rwanda)	Q1 2027	$30K-60K
India (CDSCO)	Class B Medical Device Software	MD-9 pathway	Q3 2027	$40K-80K
UK (MHRA)	Class IIa (UKCA)	UKCA conformity assessment	Q2 2027	$80K-120K

5.2 EU AI Act Compliance (August 2026 Enforcement)

Our system is classified as **high-risk AI** under EU AI Act Article 6 (medical devices under MDR). Required compliance measures and our readiness:

Requirement	Article	Our Status	Evidence
Risk management system	Art. 9	Implemented	`src/governance/audit_trail.py`, risk documentation
Data governance	Art. 10	Implemented	Dataset cards, data validation pipeline, DVC versioning
Technical documentation	Art. 11	Implemented	Model cards, 12 documentation guides
Record-keeping	Art. 12	Implemented	Prediction logging, audit trail (SHA-256 chained)
Transparency	Art. 13	Implemented	4 explainability methods, model cards
Human oversight	Art. 14	Implemented	Human-in-the-loop review queue
Accuracy, robustness, cybersecurity	Art. 15	Implemented	68 tests, drift monitoring, JWT auth, rate limiting
Quality management system	Art. 17	Partial	CI/CD pipeline, needs formal QMS documentation
Conformity assessment	Art. 43	Not started	Requires Notified Body engagement

Regulatory advantage: We are among the first ophthalmic AI systems designed from the ground up for EU AI Act compliance. Most competitors will need to retrofit compliance — we have it architecturally embedded.

5.3 FDA Strategy: De Novo Classification

We recommend the **De Novo pathway** because:
- No substantially equivalent predicate exists for 45-disease multi-label retinal screening
- De Novo creates a new classification, establishing us as the predicate for future competitors
- Expected timeline: 12-18 months from submission
- Requires prospective clinical study (300-500 patients, multi-site)

Pre-submission meeting: Target Q3 2026 to align on clinical study design and intended use statement.

5.4 Clinical Validation Roadmap

Phase	Study	Sites	Patients	Endpoint	Timeline
Retrospective	Multi-reader multi-case	3 sites	1,000 images	Sensitivity/specificity vs. expert panel	Q2-Q3 2026
Prospective (pilot)	Single-arm screening	2 sites (Uganda, UK)	500 patients	Time-to-diagnosis, referral accuracy	Q4 2026 - Q1 2027
Prospective (pivotal)	Multi-site RCT	5+ sites	2,000 patients	Sensitivity > 85%, specificity > 80% for top 10 diseases	Q1-Q3 2027
Real-world evidence	Registry study	10+ sites	Ongoing	Performance monitoring, drift detection	Post-launch

6. Go-to-Market Strategy

6.1 Phased Market Entry

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PHASE 0: CLINICAL VALIDATION          PHASE 1: EARLY ADOPTERS
(Q2-Q4 2026)                          (Q1-Q2 2027)
┌───────────────────────────┐         ┌───────────────────────────┐
│ • Retrospective study     │         │ • CE Mark obtained        │
│ • EU AI Act conformity    │         │ • 5-10 pilot sites        │
│ • Pre-Sub meeting (FDA)   │         │ • East Africa launch      │
│ • 3 anchor partnerships   │         │ • 2-3 OEM partnerships    │
│ • Hire regulatory + sales│  ───▶   │ • First revenue           │
│ • Seed/Series A ($2-4M)   │         │ • Clinical evidence       │
│                           │         │ • Conference presence     │
│ Revenue: $0               │         │ Revenue: $200K-500K       │
└───────────────────────────┘         └───────────────────────────┘
           │                                     │
           ▼                                     ▼
PHASE 2: SCALE                        PHASE 3: MARKET LEADERSHIP
(Q3 2027 - Q4 2027)                   (2028+)
┌───────────────────────────┐         ┌───────────────────────────┐
│ • FDA clearance           │         │ • 200+ active sites       │
│ • 50-100 active sites     │         │ • 5+ country approvals    │
│ • US market entry         │         │ • OEM embedded in cameras│
│ • Series B ($8-15M)       │         │ • Population screening    │
│ • Expand to 45 diseases   │  ───▶   │   programs (3+ countries)│
│   (start with top 10-15)  │         │ • Expanded modalities     │
│ • Government tenders      │         │   (OCT, slit-lamp)        │
│                           │         │ • Platform ecosystem      │
│ Revenue: $2M-5M ARR       │         │ Revenue: $15M-45M ARR     │
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6.2 Launch Markets (Priority Order)

Priority	Market	Rationale	Regulatory	Channel
1	**Uganda + East Africa**	Home market, clinical knowledge graph calibrated for local prevalence, massive specialist shortage (1 ophthalmologist per 400K people), government relationships	EAC registration	Direct + Ministry of Health
2	**United Kingdom**	Strong NHS AI adoption program (AI Diagnostic Fund), NICE health tech assessment pathway, English-speaking, UKCA pathway	UKCA (Class IIa)	NHS procurement + specialty clinics
3	**European Union**	CE Mark provides access to 27-country market, EU AI Act readiness is differentiator	CE Mark (MDR)	Distributors + hospital networks
4	**India**	1.4B population, 12M ophthalmology visits/year, rapid AI adoption (ABDM digital health infrastructure), cost-sensitive (our strength)	CDSCO Class B	OEM partnerships + Aravind-type eye hospitals
5	**United States**	Largest revenue market, but most expensive to enter (FDA + sales force) — enter after validation proof points	FDA De Novo	Enterprise sales + EHR integration

6.3 Channel Strategy

Direct Sales (Tier 1 & 2): Enterprise sales team targeting hospital networks and large specialty practices. Sales cycle: 3-6 months. Quota-carrying AEs supported by clinical specialists who can demo the explainability features and discuss clinical validation.

OEM Partnerships (Tier 4): Embed our API into fundus camera software (Topcon, Canon, Optomed, Remidio) and telehealth platforms. The camera becomes an AI-powered screening station. Revenue share model.

Government/NGO Channel (Tier 3): Respond to WHO, World Bank, and national government tenders for population screening programs. Often requires on-premise deployment (our Docker architecture supports this) and local language support.

Conference-Led Demand Generation: Presence at AAO, ARVO, EURETINA, WOC, and AIOS. Focus on peer-reviewed publications and live demos. Ophthalmology is a relationship-driven specialty — conference presence converts to pipeline.

6.4 Pricing Strategy Rationale

Our pricing is deliberately 30-50x cheaper than incumbents ($0.30-0.80 vs. $25-40/test) for three reasons:

1. **Market creation, not market share**: At $25/test, most of the world cannot afford retinal screening. At $0.50/test, screening becomes viable for primary care clinics in LMICs, school health programs, and diabetic wellness checks. We expand the market rather than fighting for incumbent share.

2. **Volume-driven economics**: Our unit compute cost is $0.02/scan. The margin improves with volume. At 10M scans/year (achievable with 2-3 government programs), our blended cost drops to $0.04/scan.

3. **Lock-in through integration**: Low per-scan pricing encourages high-volume commitments and deep workflow integration. Once we are embedded in a hospital's screening pathway, switching costs are high.

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7. Product Strategy & Roadmap

7.1 MVP Feature Prioritization (Phase 0-1)

Not all 45 diseases need to launch simultaneously. Clinical impact and regulatory efficiency demand prioritization:

Launch Set (10 diseases) — highest prevalence, strongest clinical evidence, regulatory precedent:

Disease	Code	Prevalence	Clinical Impact	Regulatory Precedent
Diabetic Retinopathy	DR	Very High	Sight-threatening	FDA-cleared predicates exist
Age-Related Macular Degeneration	ARMD	Very High	Leading cause of blindness (>50)	Multiple cleared devices
Glaucoma (Optic Disc Changes)	ODC/ODP/ODE	High	Irreversible vision loss	Emerging cleared devices
Diabetic Macular Edema	DN	High	Requires urgent treatment	FDA-cleared (with DR)
Branch Retinal Vein Occlusion	BRVO	Medium	Common vascular emergency	No cleared AI
Central Retinal Vein Occlusion	CRVO	Medium	Sight-threatening emergency	No cleared AI
Epiretinal Membrane	ERM	Medium	Common surgical indication	No cleared AI
Macular Hole	MH	Medium	Surgical emergency	No cleared AI
Hypertensive Retinopathy	HR	Medium	Systemic disease marker	No cleared AI
Retinitis Pigmentosa	RP	Low	Genetic, early detection critical	No cleared AI

Expansion Sets:
- Phase 2 (Q3 2027): Add 15 more diseases (total 25)
- Phase 3 (2028): Full 45-disease panel

7.2 Platform Evolution

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2026                    2027                    2028
2029
───────────────────────────────────────────────────────────────────────────
───
SCREENING AID           CLINICAL WORKFLOW       PLATFORM                ECO
SYSTEM
                        INTEGRATION             EXPANSION

• Single-scan           • EHR integration       • OCT support           • T
hird-party
  classification          (HL7 FHIR)             (new modality)          mo
del marketplace
• Explainability        • Automated referral    • Longitudinal          • F
ederated
  dashboard               letters                 progression             l
earning across
• Referral priority     • Bilateral analysis      tracking                s
ites
• Model card reports      (both eyes)           • Risk prediction       • R
esearch data
• Audit trail           • Patient timeline        (5-year prognosis)      c
onsortium
                        • Multi-language UI     • Edge deployment       • D
igital
                        • Offline mode            (camera-embedded)       t
herapeutics
                          (edge inference)      • Regulatory dossier      i
ntegration
                                                  automation
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8. Partnership Strategy

8.1 Strategic Partnership Framework

Partner Type	Target Partners	Value Exchange	Priority
Camera OEMs	Optomed (portable), Remidio (smartphone), Canon, Topcon	We get distribution; they get AI differentiation	Critical
Cloud / Infra	AWS HealthLake, Google Cloud Healthcare, Azure Health	We get credits + compliance infra; they get lighthouse customer	High
Academic / Clinical	Moorfields Eye Hospital, Aravind Eye Care, Mulago Hospital	We get validation data + clinical credibility; they get AI capability	Critical
EHR Vendors	Epic, Cerner (Oracle Health), OpenMRS (LMIC)	We get workflow embedding; they get AI module	High
Distributors (LMIC)	MTN MoMo Health, mPharma, Babyl (Rwanda)	We get last-mile distribution; they get clinical AI content	Medium
Regulatory	BSI (Notified Body), Emergo (FDA consulting)	We get regulatory navigation; they get a showcase client	High

8.2 Anchor Partnership Targets (Phase 0)

1. **Optomed (Finland)** — Portable fundus camera manufacturer with strong LMIC presence. Our AI + their Aurora camera = an AI-powered portable screening kit. Revenue share on bundled sales.

2. **Moorfields Eye Hospital (London)** — World-leading eye hospital with established AI research program (DeepMind collaboration). Clinical validation partner + entry point into NHS.

3. **Mulago National Referral Hospital (Kampala)** — Largest referral hospital in Uganda. Clinical knowledge graph already calibrated for Ugandan prevalence. Pilot site for East Africa launch.

9. Financial Projections

9.1 Three-Year Revenue Model

Metric	Year 1 (2027)	Year 2 (2028)	Year 3 (2029)
Tier 1 (Enterprise)	5 customers	20 customers	50 customers
Tier 1 ACV	$100K	$120K	$150K
Tier 1 Revenue	$500K	$2.4M	$7.5M
Tier 2 (Clinics)	30 clinics	120 clinics	300 clinics
Tier 2 ACV	$30K	$36K	$42K
Tier 2 Revenue	$900K	$4.3M	$12.6M
Tier 3 (Government)	2 programs	5 programs	12 programs
Tier 3 ACV	$80K	$150K	$250K
Tier 3 Revenue	$160K	$750K	$3.0M
Tier 4 (OEM)	1 partner	3 partners	6 partners
Tier 4 ACV	$200K	$300K	$400K
Tier 4 Revenue	$200K	$900K	$2.4M
Total Scans	2.1M	9.5M	32M
Total ARR	**$1.76M**	**$8.35M**	**$25.5M**
Gross Margin	68%	74%	78%

9.2 Funding Strategy

Round	Timing	Amount	Use of Funds	Key Milestones to Unlock
Pre-Seed (current)	Completed	Bootstrapped	MVP development, initial clinical data	Working product, 45-disease classification
Seed	Q3 2026	$2-4M	Regulatory (CE + FDA pre-sub), clinical validation study, first 3 hires (regulatory, sales, clinical)	CE Mark filing, retrospective study results
Series A	Q2 2027	$8-15M	FDA submission, US sales team, scale infrastructure, 3 OEM integrations	CE Mark granted, FDA pre-sub complete, first $1M ARR
Series B	Q1 2029	$25-40M	International expansion (5+ markets), platform features, 50+ person team	FDA clearance, $8M+ ARR, 3+ government programs

9.3 Investor Profile Targets

- **Seed**: Health-tech focused VCs (Khosla Ventures, 8VC, General Catalyst Health), African health-tech funds (TLcom Capital, Novastar Ventures, Future Africa)
- **Series A**: Growth health-tech VCs (Andreessen Horowitz Bio+Health, GV, Lux Capital) + strategic investors (Optomed, Topcon Ventures)
- **Series B**: Crossover funds + international health investors (IFC, Leapfrog Investments, Axa Venture Partners)

10. Intellectual Property Strategy

10.1 Patent Portfolio Plan

Patent Application	Category	Scope	Filing
Clinical knowledge graph-augmented GNN for multi-label medical image classification	Core Algorithm	The method of embedding disease co-occurrence, severity, and prevalence priors into graph neural network inference for multi-label diagnosis	Q3 2026
Adaptive multi-resolution encoding with sparse top-K attention for retinal images	Architecture	Multi-resolution patch encoding with O(n*k) attention for efficient fundus analysis	Q4 2026
Automated referral priority ranking from multi-label disease predictions	Clinical Application	Method of computing referral urgency from predicted disease combinations, severities, and clinical relationships	Q1 2027
Drift-aware model governance with immutable audit trail for SaMD	MLOps/ Regulatory	System for continuous monitoring, drift detection, and SHA-256 chained audit logging for regulatory compliance	Q1 2027

10.2 Trade Secrets

- Clinical knowledge graph edge weights and disease relationship encoding (144 relationships)
- Disease prevalence calibration data for specific populations (Uganda, East Africa)
- Optimal per-class threshold configurations for multi-label prediction
- Asymmetric loss hyperparameters tuned for extreme class imbalance in retinal diseases

10.3 Defensive IP

- Open-source the base ViGNN architecture (without clinical knowledge graph) to establish prior art and build community
- File provisional patents before any conference publications or preprints
- Consider PCT (Patent Cooperation Treaty) filing for international coverage

11. Team & Organizational Plan

11.1 Founding Team Gaps & Key Hires

Role	Priority	Why	Timing
VP Regulatory Affairs	Critical	CE Mark + FDA pathway execution, QMS establishment	Q3 2026
Clinical Director (Ophthalmologist)	Critical	Clinical validation study design, KOL relationships, medical affairs	Q3 2026
Head of Sales (Enterprise)	High	Build pipeline for Tier 1 + 2 customers	Q4 2026
ML Engineer (Production)	High	Model optimization, ONNX/edge deployment, retraining pipeline	Q3 2026
DevOps / SRE	High	Production infrastructure, SOC2, HIPAA compliance	Q4 2026
Clinical Success Manager	Medium	Post-sale clinical integration and training	Q1 2027
Regulatory Associate	Medium	Documentation, submission preparation, post-market surveillance	Q1 2027

11.2 Organizational Structure (Year 2)

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CEO / Co-Founder
├── CTO / Co-Founder
│   ├── ML Engineering (3)
│   ├── Platform Engineering (2)
│   └── DevOps / SRE (1)
├── VP Regulatory Affairs
│   ├── Regulatory Associate (1)
│   └── QA / QMS (1)
├── Clinical Director
│   ├── Clinical Success (2)
│   └── Medical Affairs (1)
├── Head of Sales
│   ├── Enterprise AEs (2)
│   └── Partnerships (1)
└── Head of Operations
    ├── Finance (1)
    └── Legal (outsourced)

Total: ~18 FTE by end of Year 2
```

12. Risk Framework

12.1 Risk Matrix

Risk	Probability	Impact	Mitigation
FDA De Novo rejection or delay	Medium	High	Engage regulatory consultant early, align on clinical study design via Pre-Sub, prepare 510(k) fallback (subset of diseases)
Clinical validation fails to meet endpoints	Low-Medium	Critical	Power study adequately, use adaptive trial design, focus on top 10 diseases first
Competitor launches 10+ disease screening	Medium	Medium	Accelerate to market, leverage LMIC-first positioning, deepen clinical knowledge graph moat
Dataset bias (demographics)	Medium	High	Multi-site validation, fairness evaluation framework already built, diversify training data through clinical partnerships
GPU cost increase / supply constraints	Low	Medium	CPU inference support already built, ONNX export enables diverse hardware, negotiate cloud commitments
Regulatory landscape changes	Medium	Medium	EU AI Act compliance already embedded, maintain regulatory counsel, participate in standards bodies
Reimbursement challenges (CPT codes)	High (US)	High (US)	Start with non-reimbursement markets (LMIC, self-pay), apply for CPT Category III code, partner with payers
Key person dependency	Medium	High	Document all systems (12 guides already written), cross-train team, vest founders over 4 years
Cybersecurity / data breach	Low	Critical	JWT auth, rate limiting, security scanning in CI, SOC2 Type II by Year 2, HIPAA BAA for US customers

12.2 Kill Criteria

The venture should be reconsidered if any of the following occur:

1. Retrospective clinical study shows sensitivity < 70% for top 5 diseases (indicates fundamental model limitation)
2. CE Mark and FDA both rejected after appeals (regulatory path blocked)

3. Three or more competitors achieve 15+ disease screening with regulatory approval before our first clearance
4. Unable to raise Seed round by Q1 2027 (market does not believe in the opportunity)
5. Zero paying customers after 12 months post-CE Mark (product-market fit failure)

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13. Implementation Roadmap: First 18 Months

Q2 2026 (Now - June 2026)

- [ ] Complete retrospective clinical validation study design
- [ ] Engage regulatory consultant for CE Mark + FDA Pre-Sub
- [ ] File first provisional patent (clinical knowledge graph GNN)
- [ ] Begin Seed fundraising ($2-4M target)
- [ ] Establish advisory board (2 ophthalmologists, 1 regulatory expert, 1 health-tech founder)
- [ ] Sign LOI with Mulago Hospital for pilot site

Q3 2026

- [ ] Hire VP Regulatory Affairs + Clinical Director + ML Engineer
- [ ] Execute retrospective study (1,000 images, 3 sites)
- [ ] Submit CE Mark technical documentation to Notified Body
- [ ] FDA Pre-Submission meeting
- [ ] Close Seed round
- [ ] Begin Optomed partnership discussions
- [ ] SOC2 Type I preparation

Q4 2026

- [ ] EU AI Act conformity assessment (August 2026 enforcement deadline)
- [ ] Prospective pilot study (500 patients, Uganda + UK)
- [ ] CE Mark expected grant
- [ ] Hire Head of Sales
- [ ] First pilot deployments (Tier 2, 3-5 clinics)
- [ ] Conference presence: AAO Annual Meeting

Q1 2027

- [ ] First revenue (Tier 2 clinics + Tier 3 pilot)
- [ ] East Africa market launch (Uganda, Kenya, Rwanda)

- [ ] FDA De Novo submission
- [ ] EHR integration development (HL7 FHIR)
- [ ] Second patent filing (referral priority ranking)
- [ ] Hire clinical success team

Q2 2027

- [ ] Series A fundraising ($8-15M)
- [ ] UK market entry (UKCA + NHS AI Diagnostic Fund application)
- [ ] First OEM partnership signed (Optomed or Remidio)
- [ ] 20+ active clinical sites
- [ ] Pivotal clinical study initiation (2,000 patients, 5+ sites)
- [ ] Conference presence: ARVO Annual Meeting

Q3-Q4 2027

- [ ] FDA clearance (target)
- [ ] US market entry preparation
- [ ] India CDSCO submission
- [ ] $2M+ ARR milestone
- [ ] 50+ active clinical sites
- [ ] Platform v2: bilateral analysis, patient timeline, multi-language

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14. Success Metrics & KPIs

Clinical

Metric	Year 1 Target	Year 3 Target
Sensitivity (top 10 diseases)	> 85%	> 90%
Specificity (top 10 diseases)	> 80%	> 85%
Referral accuracy	> 90%	> 95%
Time-to-diagnosis reduction	40%	60%
False positive rate	< 15%	< 10%

Commercial

Metric	Year 1 Target	Year 3 Target
ARR	$1.76M	$25.5M
Total scans processed	2.1M	32M
Active clinical sites	38	362
Net revenue retention	110%	130%
Customer acquisition cost (CAC)	$15K	$10K
CAC payback period	8 months	5 months
Gross margin	68%	78%

Operational

Metric	Target
System uptime	99.9%
Inference latency (p99)	< 100ms
Drift detection response time	< 24 hours
Model retraining cadence	Quarterly
Regulatory audit readiness	Always

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15. Strategic Summary

Why This Wins

1. **Right problem, right time**: 2.2B people with vision impairment, <250K ophthalmologists globally, EU AI Act creating regulatory moats for compliant systems, AI reimbursement codes emerging.

2. **Structural cost advantage**: 25ms inference = $0.02/scan compute cost. Competitors charging $25-40/test are 50-100x our cost basis. We can profitably serve markets they cannot.

3. **Technical moat**: Clinical knowledge graph with 144 disease relationships is not a model weight — it is structured clinical knowledge that takes years of ophthalmological expertise to build and validate. Combined with 45-disease multi-label classification, this is defensible.

4. **LMIC-first is a strategy, not charity**: East Africa has the worst ophthalmologist shortage and the least AI competition. We build clinical evidence, refine the product, and generate revenue in markets where we are the only option — then enter the US/EU with validated technology and real-world evidence that no competitor has.

5. **Regulatory-first architecture**: EU AI Act compliance is baked in, not bolted on. As regulations tighten globally, our head start becomes a compounding advantage.

The Ask

$2-4M Seed round to execute Phase 0 (clinical validation + regulatory submission) and Phase 1 (first revenue), positioning for a $8-15M Series A at CE Mark + first $1M ARR milestone.

This strategy is a living document. Review quarterly and update based on clinical validation results, regulatory feedback, and market signals.